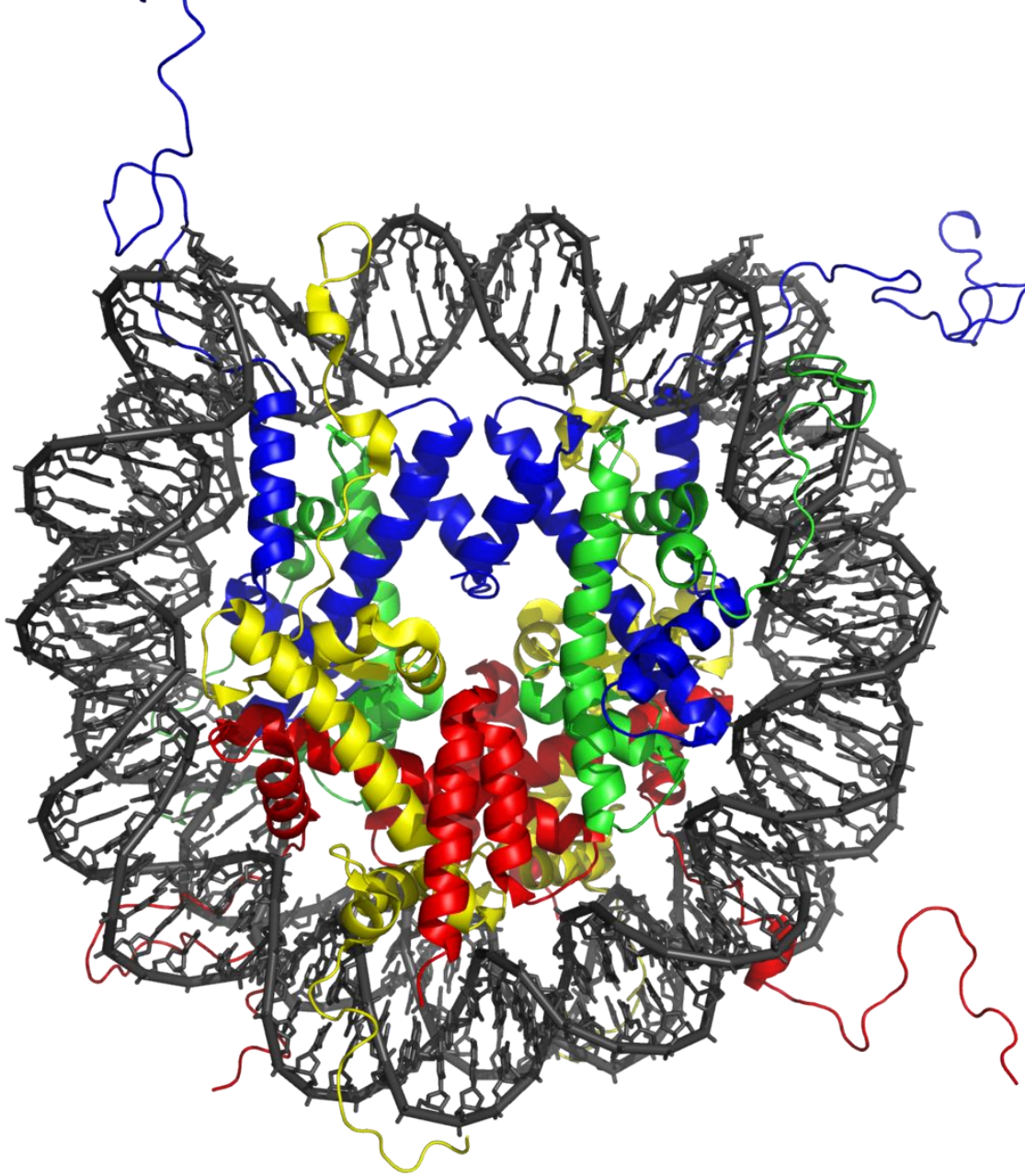
DP Biology – A1.2 (HL extension)

DNA: direction, pairing and packing

- How do nucleic acids store information that can be inherited?
- Which features of DNA allow it to be copied accurately?



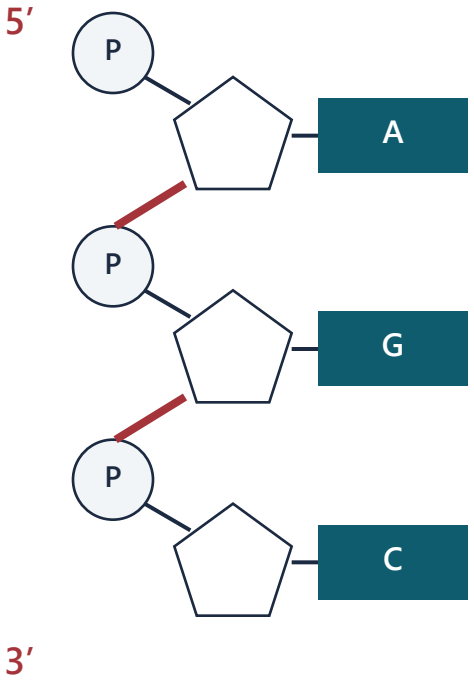
HL statements A1.2.11–15. Textbook pp. 25–33.
Image: a nucleosome, DNA wrapped around eight histones. Zephyris, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	A1.2.11–12	Explain the 5′ to 3′ direction of nucleic acid strands, and why purine-pyrimidine pairing keeps the helix regular
2	A1.2.13	Describe the structure of a nucleosome, and explain why DNA is packed this way
3	A1.2.14–15	Evaluate the evidence from Hershey and Chase, and from Chargaff's data, for the nature of the genetic material

Every strand has a direction

A1.2.11



Phosphate (5′) to sugar (3′) links

The carbons in deoxyribose (and ribose) are numbered 1′ to 5′. The base is on carbon 1′ and the phosphate on carbon 5′.

In a strand, each phosphate links **carbon 3′** of one sugar to **carbon 5′** of the next. So one end has a free phosphate (the **5′ end**) and the other a free -OH on carbon 3′ (the **3′ end**).

The two strands of DNA run in opposite directions: they are **antiparallel**. RNA strands have the same 5′ to 3′ direction.

Key idea

New nucleotides can only be added to the free 3′ -OH of a growing strand, so nucleic acids are always built in the 5′ to 3′ direction.

Why direction matters

A1.2.11

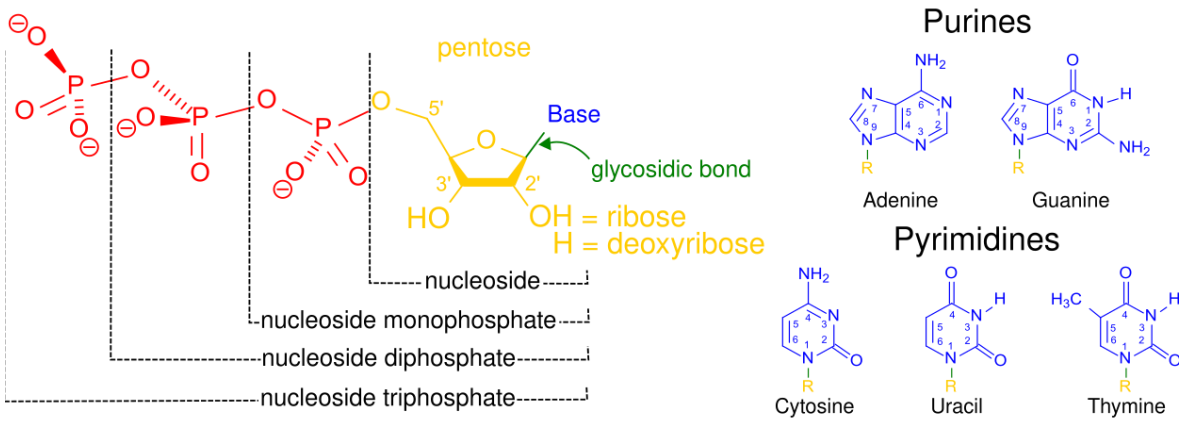
Process	Role of directionality
Replication	DNA polymerase adds nucleotides only to the 3′ end, so new strands grow 5′ to 3′. Because the strands are antiparallel, they are copied in opposite directions
Transcription	RNA polymerase reads the template strand 3′ to 5′ and builds mRNA 5′ to 3′
Translation	Ribosomes read mRNA from the 5′ end towards the 3′ end, so codons are read in a fixed order

Common mistake

The template strand for transcription is the non-coding strand. The coding strand has the same sequence as the mRNA (with T in place of U).

Purines pair with pyrimidines

A1.2.12



Nucleotide parts, purines and pyrimidines. Public domain.

	Purines	Pyrimidines
Rings	Two	One
Bases	Adenine, guanine	Cytosine, thymine (uracil in RNA)

Each base pair, A-T or C-G, is always **one purine plus one pyrimidine**. Both kinds of pair are the **same width**, so the double helix has the same diameter whatever the sequence.

Two purines together would bulge; two pyrimidines would leave a gap. Neither would fit.

A-T pairs have **two** hydrogen bonds; C-G pairs have **three**.

So

Any base sequence can be stored without changing the shape of the molecule, and enzymes can work along all of it.

Working with data: C-G pairs and stability

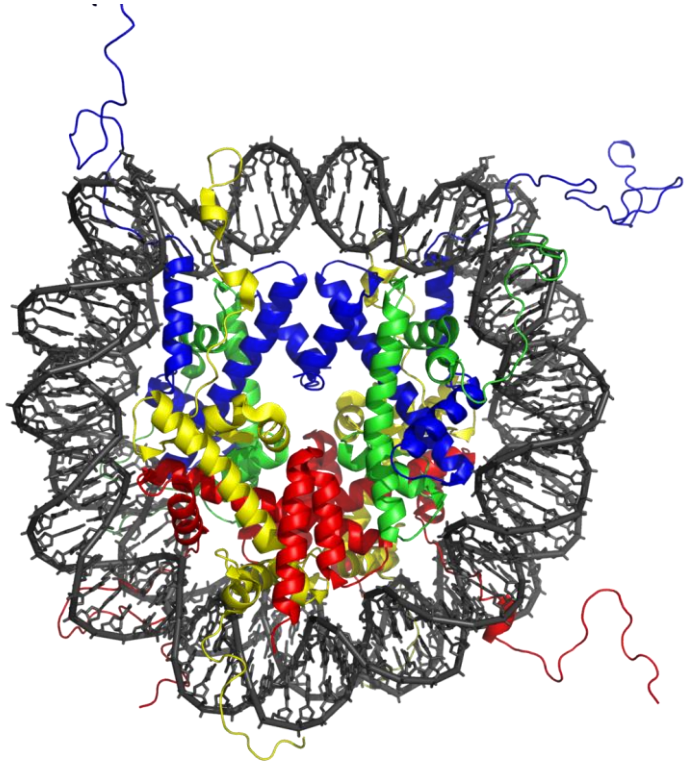
A1.2.12

Heating separates the two strands of DNA. The temperature at which half the DNA has separated is its melting temperature. Values calculated from the published relationship of Marmur and Doty (1962), for DNA in a standard salt solution.

C-G pairs / %	Melting temperature / °C
30	82
40	86
50	90
60	94
70	98

1. Describe the relationship between C-G content and melting temperature.
2. Explain the relationship using hydrogen bonds.
3. Predict the melting temperature of DNA with 45% C-G pairs.
4. Bacteria that live in hot springs often have DNA with a high C-G content. Suggest why.

The nucleosome



In eukaryotes, DNA is wound around proteins called **histones**. Each **nucleosome** has a core of **eight histones** (two each of four types), with DNA wrapped about twice around it.

Linker DNA joins one nucleosome to the next, and another histone, **H1**, binds where the DNA enters and leaves, holding it in place.

Histones are rich in positively charged amino acids (lysine and arginine), which attract the negatively charged phosphates of DNA.

Nucleosome (1KX5): DNA in grey around histones. Zephyris, CC BY-SA 3.0.

Eukaryotic DNA looks like beads on a string. Prokaryotic DNA has no histones: it is "naked".

Why package DNA?

A1.2.13

Fact	Size
Total DNA in one human nucleus	About 2 m long
Nucleus	About 5 to 10 μm across
Chromosome at metaphase	About 5 μm long, holding about 5 cm of DNA

Packing lets 2 m of DNA fit into the nucleus.

Highly condensed chromosomes can be moved in cell division without tangling or breaking.

Unwinding particular regions lets genes be transcribed, so packing helps **control gene expression**.

Molecular visualization

Open a nucleosome structure (for example PDB 1KX5) in a 3D viewer. Find the two copies of each histone and colour the positive amino acids. Where are they, compared with the DNA?

Hershey and Chase, 1952

A1.2.14



Martha Chase and Alfred Hershey. CC BY-SA 4.0.

Phages with sulfur-35 (protein labelled)	Phages with phosphorus-32 (DNA labelled)

1. Phages infect E. coli bacteria
2. A blender shakes the phage coats off the cells
3. Centrifuge: cells form a pellet; coats stay in the liquid

Bacteriophages are viruses made only of DNA inside a protein coat. Proteins contain sulfur but almost no phosphorus; DNA contains phosphorus but no sulfur. So Hershey and Chase could label the protein and the DNA separately with radioactive isotopes and follow where each went.

What Hershey and Chase found

A1.2.14

Result	Sulfur-35 batch (protein)	Phosphorus-32 batch (DNA)
Radioactivity in pellet (bacteria)	Very little	Most
Radioactivity in liquid (phage coats)	Most	Little
Label in the new phages made	None	Some

Only the phage **DNA** entered the bacteria; the protein coats stayed outside. Yet the infected bacteria went on to make complete new phages, coats and all. So DNA, not protein, carried the genetic information.

Technology

The experiment depended on radioactive isotopes, which became widely available after the Second World War.

Evaluating the evidence

A1.2.14

Why protein was expected

Before 1952, many scientists thought genes were made of protein. Proteins are built from 20 amino acids, so they seemed to offer far more variety than DNA, with only four bases.

Not 100% clean

Some sulfur-35 ended up in the pellet, and some phosphorus-32 stayed in the liquid: blending did not remove every coat, and some phages had not yet injected their DNA. The difference was still large and consistent, so the conclusion was accepted.

Discuss

One experiment rarely settles a question on its own. How did the Hershey-Chase result fit with Avery's earlier work (1944), which showed that purified DNA could transform bacteria? Why is agreement between different methods convincing?

Chargaff's data on base composition

Source of DNA	A %	G %	C %	T %
Human	30.9	19.9	19.8	29.4
Wheat	27.3	22.7	22.8	27.1
E. coli	24.7	26.0	25.7	23.6
Yeast	31.3	18.7	17.1	32.9

In every species, **A = T** and **G = C**, so purines = pyrimidines.

But the proportion of A + T compared with G + C **varies between species**.

Data from Erwin Chargaff and colleagues, about 1950 (published values, rounded).

Try it

Calculate (A + G) / (C + T) for each species. What do you notice, and how does base pairing explain it?

How Chargaff's data falsified a hypothesis

A1.2.15

In the 1930s and 1940s, the **tetranucleotide hypothesis** proposed that DNA was a simple repeat of the four bases in equal amounts (such as ACGT ACGT ACGT). If so, DNA could not carry information, and genes would have to be protein.

Chargaff showed that the four bases are **not** present in equal amounts, and that the proportions **differ between species**. This **falsified** the tetranucleotide hypothesis.

The equal amounts of A and T, and of G and C, were later explained by Watson and Crick's model of complementary base pairing (1953).

Nature of science

A single clear contradiction can falsify a hypothesis, but no number of agreeing results can prove one. Chargaff's data did not tell anyone what DNA looked like, but they ruled out one model and gave a clue to the right one.

Check-in: HL nucleic acids

A1.2.11–15

1. State which end of a DNA strand new nucleotides are added to.
2. Explain why purine-to-pyrimidine pairing keeps the double helix the same width.
3. Describe the structure of a nucleosome.
4. Explain why the phosphorus-32 label ended up inside the bacteria in the Hershey-Chase experiment.
5. Explain how Chargaff's data falsified the tetranucleotide hypothesis.

How to use this

Write full sentences, then mark with a partner.

Common mistakes to avoid

Not this	But this
New nucleotides are added to the 5′ end.	They are added to the free 3′ end; strands grow 5′ to 3′.
Histones are made of DNA.	Histones are proteins; DNA wraps around them.
Hershey and Chase showed DNA is a double helix.	They showed DNA, not protein, is the genetic material.
Chargaff showed A = G and C = T.	He showed A = T and G = C.
Prokaryotic DNA is wrapped around histones.	Prokaryotic DNA is naked: it has no histones.

Check that you can...

1. **Explain** the 5′ and 3′ ends of a strand and why strands are antiparallel
2. **Outline** how directionality affects replication, transcription and translation
3. **Explain** why purine-pyrimidine pairing gives a regular helix, and **analyse** data on C-G content
4. **Describe** the structure of a nucleosome, and **explain** why DNA is packaged
5. **Evaluate** the Hershey-Chase experiment as evidence that DNA is the genetic material
6. **Analyse** Chargaff's data, and **explain** how it falsified the tetranucleotide hypothesis

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State which end of a DNA strand new nucleotides are added to.
2. Describe the structure of a nucleosome.
3. Explain why the phosphorus-32 label ended up inside the bacteria in the Hershey-Chase experiment.

Image credits

Nucleosome 1KX5 colour coded.png – Zephyris at English Wikipedia – CC BY-SA 3.0

Nucleotides 1.svg – Boris (PNG), SVG by Sjef – Public domain

Scientist Alfred Hershey and Martha Chase.jpg – Rohit Kumar Sengupta – CC BY-SA 4.0

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